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SCHOOL OF SCIENCE AND ENGINEERING

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Requirements III — Functional, Non-functional and Domain

Three kinds of requirement, why the classification decides the architecture, and how to turn an adjective into something a student can measure on a Saturday

Session 9 of 28 · Tuesday 6 October 2026 · 1:00 – 2:20 PM

· Prof. Guillaume Sanje Mpacko, Ph.D.

Requirements III — Functional, Non-functional and Domain

SESSION 9 OF 28

Tuesday, 6 October 2026 · 1:00 – 2:20 PM · Unit 2 · Requirements

LEARNING OBJECTIVES

- ✓ Classify a requirement as functional, non-functional or domain, and explain why the classification changes the architecture.
- ✓ State, for each of the main quality attributes, the question it asks of SANTE-BASSAM and the mechanism chosen to satisfy it.
- ✓ Write a non-functional requirement with a measurable fit criterion — scale, meter, must, plan, source, rationale.
- ✓ Trace every requirement back to an observation row, an interview question, a regulation clause or a prototype session.

FROM SESSIONS 7 AND 8

You now have evidence: a stakeholder map, an elicitation plan, three-column records from the desk and from outreach, and two prototype sessions. What you do not have is a **specification** — anything a person could

build to, or test against. Today converts the evidence into requirements, under one rule: **every requirement carries its source**.

TEAM PROJECT · 0

first 15 minutes of class

The records come back — turning rows into candidate requirements

Applying Session 8. Three pairs went out with three instruments. Today the rows land on the table and the team decides what they change.

WHAT HAPPENS

1. **Each pair reads out its two strongest rows** — strongest meaning the ones carrying a count. Observation and inference only; the rest of the team supplies the test column. **Six minutes.**
2. **Three decisions, out loud.** For each strong row: does it *confirm* an existing story, *change* one, or *create* a new one? Write the verdict against the story number. **Six minutes.**
3. **Kill something.** Name at least one backlog hypothesis the field work **refuted**, and move it out. A fortnight of elicitation that removes nothing was not elicitation. **Three minutes.**

Output: in the progression log — the rows, the confirm / change / create verdict for each, and the story you deleted with the row number that killed it.

Next session's slot: reconciling the backlog against the requirement list, so bring both.

1 · The asymmetry that makes this session matter

Functional requirements decide **what** you build. Quality attributes decide **how you structure it**. Those two are not symmetrical, and the difference has a practical consequence you have already lived through on this project.

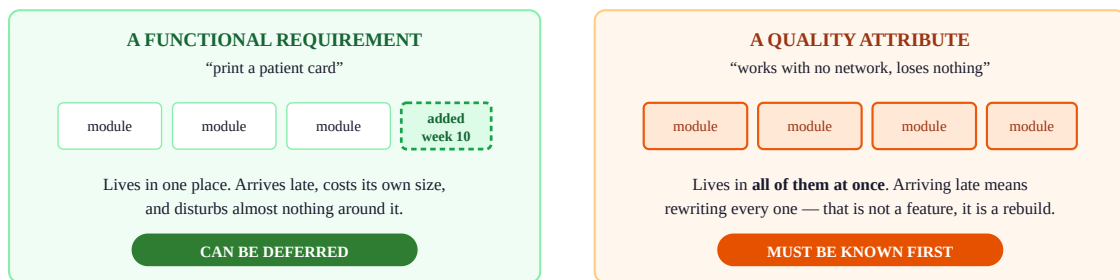


Figure 1. The architectural asymmetry, and the reason non-functional requirements belong in the specification before the first design meeting. Your team already relied on this: offline-first was decided in the sprint 0 spike, not in week 10, because it could not have been added in week 10.

2 · Three kinds of requirement

Functional · FR

What the system must do.

Services, inputs, outputs, behaviour under stated conditions.

Verified by a test that passes or fails.

Non-functional · NFR

How well, and under what constraints. Performance, availability, security, usability, maintainability.

Verified against a fit criterion — a measurement and a threshold.

Domain · DR

Imposed by the world, not wished for by anyone.

Usually legal, usually non-negotiable, usually the ones nobody thinks to state.

Verified by inspection against the source document.

Domain requirements are the ones that bite

Nobody will ever tell you a domain requirement in an interview, because to them it is not a requirement — it is simply how the world is. You find them by reading the regulation and the guidelines. A student team that skips document analysis discovers DR-01 in week 12, when the Ministry rejects its identifiers.

APPLIED TO SANTE-BASSAM

Twenty-four requirements — and not one of them without a source

FUNCTIONAL — TWELVE, FROM THE BACKLOG AND THE ELICITATION RECORDS

ID	The system shall...	Source · story
FR-01	register a new patient with given name, family name, birth date (year only is acceptable), sex and village; optionally the mother's name and a phone number	card fields; obs. 4 · S-04
FR-02	find patients by card number, by name, or by mother's name and village — ranked by similarity, with village and birth year shown on each row	obs. 4 (19 of 31); prototype A · S-05
FR-03	warn the clerk when a new registration matches an existing patient on family name, mother, village and birth year	obs. 4; risk R7 · S-15
FR-04	store every registration and visit locally before displaying any confirmation	nurse Q8; obs. 7 · S-01
FR-05	mark offline records pending, upload them in creation order, and mark them synced only on server acknowledgement	sprint 0 spike; CHW shadowing · S-07, S-08
FR-06	flag any record failing upload three times for supervisor review, and retain it on the device	nurse Q10 · S-09
FR-07	book an appointment on a date, and reschedule it while keeping the history	appointment book · S-10
FR-08	record each appointment's outcome — attended, missed or rescheduled — with an optional follow-up date	nurse Q5; obs. 3 · S-11
FR-09	list every patient whose follow-up falls in the current week, by village, naming the CHW where there is no phone	nurse Q4–Q5; prototype B · S-12
FR-10	produce a monthly report of visits by village and outcome, exportable as a spreadsheet	officer; obs. 5 · S-13, S-14
FR-11	accept a village only from the controlled list of 23, with type-ahead — never free text	obs. 5; rejected reports · new S-18
FR-12	make each visit outcome visible to the pharmacy on the same clinic day	pharmacy assistant, wk 4 · new S-17

NON-FUNCTIONAL — EIGHT HEADLINE STATEMENTS (FIT CRITERIA IN §4)

ID	Attribute	Statement	Source
NFR-01	Availability (offline)	Every function the clerk and CHW use in a clinic day works with no network.	CHW shadowing; obs. 2
NFR-02	Reliability (power)	No committed record is lost or left partial when power fails at any moment.	nurse Q8; obs. 7
NFR-03	Performance	The first page of a search over 10,000 records appears within two seconds on the desk laptop.	queue observed at desk
NFR-04	Usability	A trained clerk registers a patient no slower than she does on paper.	obs. 6 — 75 s median
NFR-05	Auditability	Who recorded or changed any record, and when, can be reconstructed.	Ministry §4; officer
NFR-06	Security	Data on either device is unreadable without the device passcode and an application login.	regulation
NFR-07	Maintainability	A student who did not write it can add a new Patient field end to end in under a day.	the empty chair; turnover
NFR-08	Reliability (battery)	The tablet application leaves at least eight hours of outreach battery life.	CHW shadowing

DOMAIN — FOUR, FROM THE REGULATION AND THE REGISTER GUIDELINES, FROM NO INTERVIEW AT ALL

DR-01

Identifiers shown to the patient or to the Ministry conform to the national health-ID format; the internal identifier maps one-to-one to it once assigned.

DR-02

Records of patients under 18 record the consenting guardian; no disclosure outside the post without that consent.

DR-03

Visit records are retained for the statutory period. They are never deleted — only marked void, with a reason.

DR-04

The monthly report contains exactly the columns of the Ministry template, in its order.

What the classification bought

FR-11 and FR-12 are new since Session 5 — one from an observation row, one from a stakeholder found by a referral question. NFR-01 and NFR-02 are the reason the architecture is offline-first with a journaled store. And DR-03 will shortly overrule a stakeholder wish, which is the most interesting thing that happens in Session 10.

3 · Quality attributes: the question each one asks

A quality attribute is not a property you sprinkle on at the end. Each one asks a specific question of your system, and each is satisfied by a specific **mechanism** that lives somewhere definite in the architecture. Knowing where it lives is what tells you whether it can be added later.

Attribute	The question it asks of SANTE-BASSAM	Mechanism chosen	Where it lives
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Availability	Can the clerk work through a whole clinic day with no network?	Local-first store; sync is a background concern, never a precondition	everywhere
Reliability	What survives a power cut mid-write?	Journalled writes; commit before confirming	everywhere
Performance	How long does the queue wait for a search?	Prefix index over names and mothers	one module
Usability	Is it faster than the paper she already has?	Field order copied from the card; type-ahead on village	the screens
Security	What happens when the tablet is left in a taxi?	Device passcode, application login, encryption at rest	a layer
Auditability	Can we say who changed this record, and when?	Append-only change log; void, never delete	the data model
Maintainability	Can the next student add a field without archaeology?	One data dictionary; field definitions in one place	structure
Portability	Laptop and tablet from one codebase?	Shared core, thin platform shells	everywhere
Interoperability	Will the Ministry accept our export?	Template columns generated from DR-04, not hand-built	one module
Scalability	6,000 patients now. What about 60,000?	Explicitly out of scope, in writing, this semester	—

Read the last column. Everything marked everywhere had to be settled before the first line of code; everything marked one module could safely wait. That single column is the whole argument of §1, made concrete.

4 • Fit criteria: turning an adjective into a test

“The system shall be fast” is not a requirement, because no two people would test it the same way. A fit criterion is what converts a quality into something a student could measure on a Saturday. Six lines, and every one of them is load-bearing.

THE SIX LINES — PLANGUAGE

SCALE

THE UNIT

What exactly is measured, in what units. *Seconds from key press to first result row.*

METER

THE PROCEDURE

How it is measured, on which device, by whom. This is the line students skip and the line that decides whether the criterion is real.

MUST

THE THRESHOLD

Below this the system has failed. Not an aspiration — a failure line.

PLAN

THE TARGET

What you are aiming for, which is better than MUST.

SOURCE

WHERE THE NUMBER CAME FROM

An observation row, an interview, a regulation — or an honest admission that it is a guess to be measured.

RATIONALE

WHY THIS NUMBER

What goes wrong at the MUST level. If you cannot say, the number is arbitrary.

APPLIED TO SANTE-BASSAM

Four of the eight, written out in full

NFR-03 · PERFORMANCE — SEARCH LATENCY

ATTRIBUTE : time from the clerk's last keystroke to the first result row
SCALE : seconds, 95th percentile over 100 searches
METER : scripted search of 100 random real surnames against a
10,000-record database, on the actual desk laptop, cold start
MUST : ≤ 2.0 s
PLAN : ≤ 0.8 s
SOURCE : obs. row 6 -- the queue grows when the clerk waits
RATIONALE : above 2 s she starts writing on paper "to catch up later",
which is the behaviour that creates the partial records

NFR-04 · USABILITY — REGISTRATION SPEED

ATTRIBUTE : time for a trained clerk to register a new patient
SCALE : seconds, median over 10 consecutive registrations
METER : the actual clerk, at the desk, real patients, stopwatch;
"trained" = after one 20-minute walkthrough and one day of use
MUST : ≤ 75 s
PLAN : ≤ 50 s
SOURCE : obs. row 6 -- 75 s is her median on paper today
RATIONALE : slower than paper and she will keep the paper; then we have
two registers and neither is complete

NFR-02 · RELIABILITY — POWER FAILURE

ATTRIBUTE : records lost or left partial when power fails during a write
SCALE : count of partial or lost records out of 20 induced failures
METER : pull the laptop power at a random point during registration,
20 times; inspect the store after each
MUST : 0 of 20
PLAN : 0 of 20, and the half-typed form is recoverable on restart
SOURCE : nurse Q8 ("twice I forgot to copy it back"); obs. row 7
RATIONALE : a lost record is a patient with no history at the next visit;
residual risk (device destroyed before sync) is named in the
SRS and accepted by the nurse in writing

NFR-08 · RELIABILITY — TABLET BATTERY

ATTRIBUTE : battery life remaining after a full outreach day
SCALE : hours of app use before the device drops below 15%
METER : tablet at 100% at 07:30, screen on for registration in bursts

as logged during CHW shadowing, no charging, GPS off

MUST : ≥ 8 h

PLAN : ≥ 10 h

SOURCE : CHW shadowing, week 4 (07:30-16:00, no charging available)

RATIONALE : a tablet that dies at 14:00 sends the CHW back to paper for the last village -- and that paper is never synced

THE TWO TESTS TO APPLY TO ANYONE'S FIT CRITERION, INCLUDING YOUR OWN

Is the METER actually runnable? Could a student, on a Saturday, with the equipment that exists, carry out that procedure and get a number? If not, it is prose. **Is the MUST an adjective in disguise?** "MUST: acceptable response time" has a threshold-shaped hole where the threshold should be.

SESSION 9 TAKEAWAY

Functional requirements say what, quality attributes say how well — and because quality attributes live in every module at once, they have to be known before the architecture, not after. Domain requirements come from the regulation and from nobody's mouth. Then the discipline that holds it all together: a requirement with no source is a guess with an identifier, and a quality with no fit criterion is a wish.

5 • In-class exercise • write the missing fit criterion

Twelve minutes in three pairs, then five minutes attacking each other's. Four of the eight NFRs are written to the standard in §4. Write one of the others.

Your pair takes one

Pair A

NFR-01 — availability (offline)

Pair B

NFR-05 — auditability

Pair C

NFR-07 — maintainability

1. **All six lines.** The METER must name the device and describe a procedure a student could run on a Saturday.
2. **The number must come from somewhere.** Cite a Session 8 observation row or interview question for the MUST — or say honestly that it is a guess to be measured in week 5.
3. **Swap and attack.** Exchange with another pair. Two questions only: *is their METER actually runnable?* and *is their MUST an adjective in disguise?*

WHERE EACH ONE IS HARD

NFR-07 is the difficult one: “a student who did not write it adds a field in under a day” needs a METER that names *which* student, *which* field, and who starts the clock. **NFR-05** tempts you into describing a log format instead of a measurement. **NFR-01** tempts you into writing a mechanism — “uses a local database” — where a scale belongs.

6 · Homework · individual · due before Session 10 (Friday 9 October)

TEMPLATE

Download `CSC4305_HW09_Template.docx` from `CSC_4305_SWE-Assignments`, fill it in, rename it `CSC4305_HW09_FirstName_LastName.docx` and commit it to `/homework/S09/` in your own private repository.

A .docx file. Not a PDF, not a link.

Q1 Classify ten statements as FR, NFR, DR or “not a requirement”. One line of reason each.

(a) The clerk can register a patient offline. (b) The system uses SQLite. (c) Reports must use the Ministry column order. (d) Search is fast. (e) A record is never deleted. (f) The tablet app should look modern. (g) A CHW can see the

patients registered by another CHW after sync. (h) Under-18 records name a guardian. (i) The team will use GitHub Actions. (j) Two clerks can use the laptop on different logins.

Q2 Write one fit criterion in full — the one your pair did *not* take in class. All six lines, your own work, not your pair's.

If the SOURCE is honestly a guess, write that. A guess labelled as a guess is good engineering; a guess labelled as a measurement is not.

Q3 Attack a fit criterion. Take the one another pair wrote in class and apply the two tests from §4. Say specifically what you would change, and why. Three or four sentences.

Name the defect, not the person. “The METER does not say which device” is useful; “this is vague” is not.

Q4 Find the source. Pick any three requirements from the tables in §2 and trace each back: which observation row, interview question, regulation clause or prototype session? Then pick one that you think has the *weakest* evidence, and say what you would do to strengthen it.

Q5 One architectural consequence. Choose one NFR and describe, in three or four sentences, what would have to be rewritten if the team had discovered it in week 12 instead of week 4. Be concrete about which parts.

This is §1, applied to your own code. It is the question the midterm will ask in some form.

Q6 Reading. Sommerville, *Software Engineering* 10th ed., §4.3. Optional: Gilb, *Competitive Engineering*, chapter 2, for Planguage.

Where this goes next. Twenty-four requirements exist and eight of them now have measurable criteria. Friday is quality control on the sentences themselves — the six properties every requirement must pass, the words that betray a defect, and the dictionary that stops the same argument recurring for the next fourteen weeks.

Key terms

functional requirement

non-functional requirement

domain requirement

quality attribute

architectural asymmetry

mechanism

fit criterion

Planguage

scale · meter · must · plan

source

rationale

Readings and references

- Sommerville, I. *Software Engineering*, 10th ed. — Chapter 4, §4.3–4.4.
- Gilb, T. *Competitive Engineering*, 2005 — Planguage.
- Robertson, S. & Robertson, J. *Mastering the Requirements Process*, 3rd ed. — fit criteria and the Volere template.
- Bass, L., Clements, P. & Kazman, R. *Software Architecture in Practice*, 4th ed. — quality attributes and tactics.